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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

COMMON TO B.E – ECE / CSE & B.TECH - IT

20EC211 - EMBEDDED SYSTEMS AND INTERNET OF THINGS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1 What is the purpose of an infinite loop in embedded programming?**
 - a) To execute a block of code once and then exit
 - b) To continuously execute a block of code until a specific condition is met
 - c) To create a delay in the program's execution
 - d) To handle interrupts and exceptions
2. **CO1 What is the primary purpose of a memory test?**
 - a) To verify the speed of the memory
 - b) To check for errors in the memory
 - c) To determine the capacity of the memory
 - d) To measure the power consumption of the memory
3. **CO3 What is the purpose of the Linux Kernel's scheduler?**
 - a) To manage memory
 - b) To handle interrupts
 - c) To schedule tasks for execution
 - d) To control input/output devices
4. **CO3 What is the purpose of a device tree in Embedded Linux?**
 - a) To describe the hardware configuration
 - b) To manage the file system
 - c) To handle network communication
 - d) To schedule tasks
5. **CO5 Which of the following is NOT a key component of an IoT system?**
 - a) Sensors
 - b) Actuators
 - c) Cloud computing
 - d) Operating system

6. **CO5** Which wireless communication technology is best suited for long-range, low-power IoT applications?
- a) Wi-Fi b) Bluetooth c) Zigbee d) LoRaWAN
7. **CO4** What is a mailbox in the context of inter-task communication?
- a) A hardware device for storing messages
- b) A software buffer for storing messages
- c) A network protocol for message transmission
- d) A memory location for sharing data between tasks

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO4** _____ occurs when a high-priority task is blocked by a low-priority task, leading to decreased system performance and potential missed deadlines.
9. **CO1** The role of a _____ in embedded systems development to compile code for the target embedded system.
10. **CO6** PaaS stands for _____.

PART - B

(5x 3 = 15 Marks)

11. **CO2** What is the role of MISRA C coding standards in embedded systems?
12. **CO6** What are the primary sensors used in an air quality monitoring system?
13. **CO1** What is the purpose of a bootloader in an embedded system?
14. **CO4** What is the difference between a hard real-time system and a soft real-time system?
15. **CO5** Name the three main service models of cloud computing.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO4** i) Explain the different types of real-time scheduling algorithms and their suitability for different applications. (8)
- CO4** ii) Describe the techniques for preventing and detecting deadlocks in real-time systems. (7)
17. **CO3** i) Explain the concept of a Linux kernel module and its lifecycle. (8)
- CO3** ii) Explain the role of the Linux kernel in handling interrupts and exceptions. (7)

18. **CO1** i) Describe the role of embedded software in an embedded system. Discuss the key considerations in embedded software development. (7)
- CO1** ii) Discuss the key steps involved in the compilation process, from source code to executable. (8)

19. **CO4** i) Explain the Banker's Algorithm with an example of a system with 5 processes (P0, P1, P2, P3, P4) and 3 resource types (A, B, C). (8)

Process	Maximum Demand (A,B,C)	Allocation (A,B,C)
P0	(7,5,3)	(0,0,0)
P1	(3,2,2)	(2,0,0)
P2	(9,0,2)	(3,0,2)
P3	(2,2,2)	(2,1,1)

How will you ensure that the system remains.

- CO4** ii) Discuss the concept of semaphores in real-time systems. Explain how semaphores can be used to implement mutual exclusion, synchronization, and resource allocation. Provide examples of real-time systems where semaphores are employed. (7)
20. **CO5** i) Describe the various wireless communication technologies used in IoT and their suitability for different applications. (7)
- CO5** ii) Discuss the key components of an IoT system and their interactions. (8)
21. **CO6** i) Design an air quality monitoring system for a city. Discuss the ethical implications of deploying air quality monitoring systems. (8)
- CO6** ii) Design a landslide detection system using IoT technology. Discuss the ethical implications of deploying landslide detection systems. (7)
22. **CO6** i) Explain the basic principle of a motion detector. What are the key components of a smart motion detector? How can IoT technology be integrated into a motion detector? Design a smart motion detector system for a home security application. (8)
- CO2** ii) Explain the MISRA C Compliance and Its Importance in Critical Systems also give the Static Code Analysis Tools for MISRA C and Dynamic Testing and MISRA C. (7)
